DDTA - Working Metamodel - Decision

NON CANONICAL WORKING DRAFT - REVISION 7

Revision 7 is a layering correction. It keeps the Decision-specific semantic core and the hierarchy/corpus invariants introduced by revision 6, but removes representation, realization, semantic-authority and authoring-review rules from the Decision-specific closed set. Those rules remain part of the study in the separate cross-cutting model-layering foundation document.

1 Purpose and evidence boundary

This study asks what a governed **Decision** means immediately below a Macro Requirement. Construction evidence remains the sixteen historical ThreatForge ADRs of MR-0001/MR-0002, followed by the independent facial-access example. ThreatForge MR-0003, MR-0004 and MR-0005 remain sequential holdouts and are not opened by this revision.

Revision 6 added two genuine cross-document semantic invariants after the independent example exposed ambiguous parentage and redundant Decision contribution. The subsequent ThreatForge regression exposed a different problem in the study itself: a realization principle (executable projections) was being applied as if it were semantic truth of every governed DDTA project. Revision 7 corrects that layering without changing the meaning of Decision.

2 Semantic position below Macro Requirement

```
Macro Requirement
    major governed concern/result
         |
         v
Decision
    significant chosen narrowing of that concern
         |
         v
Requirement-level semantics (to be studied next)
    independently checkable obligations / downstream effects
```

The last level is intentionally not frozen here. Requirement remains the working name of the next document family.

3 Decision-specific semantic rules CLOSED for the construction candidate

D-CLOSED-01 - Purpose of Decision

> **A Decision records one significant choice that resolves an unresolved problem or tension within one parent Macro Requirement and narrows that macro concern.**

Its subject may be a method, policy, convention, operating practice, semantic choice, representation choice, technology or architecture when that is the actual choice being governed.

D-CLOSED-02 - Minimal semantic core

```
Decision
|- title
|- context
|- decision
`- consequences
```

All four Decision-specific semantic attributes are required and non-null in the current candidate. Parentage is an explicit semantic relation to the Macro Requirement rather than duplicated prose.

D-CLOSED-03 - Decision-local Context

> **Context describes the unresolved local decision problem, tension or ambiguity that makes this specific choice necessary inside the already-governed parent concern.**

If Context mostly explains why the parent MR exists, it is too high-level. Context may preserve historically true circumstances of the choice, but it must not become a cumulative copy of inherited knowledge.

D-CLOSED-04 - One coherent chosen position

> **Decision states one coherent chosen position that can meaningfully be accepted, rejected or superseded as a conceptual unit.**

Split diagnostic: if A can change while B remains valid and both have independently meaningful trade-offs, the candidate likely contains more than one Decision.

D-CLOSED-05 - Consequence awareness

> **Consequences record relevant effects and trade-offs accepted because of the choice.**

Benefit, Cost, Risk and Constraint remain useful authoring labels but are not frozen semantic subtypes.

D-CLOSED-07 - No mandatory embedded Alternatives or Rationale fields

The evidence does not justify mandatory `alternatives[]` or a separate `rationale` semantic field. Substantial alternatives may exist as separately governed candidate Decisions. Reasoning is expected to be recoverable from Context + Decision + Consequences unless a future counterexample demonstrates otherwise.

4 Cross-document Decision-set invariants CLOSED

These invariants belong to the MR–Decision relation and to the governed Decision set. They are not additional fields inside a Decision.

H-CLOSED-01 - Unique semantic ownership

> **Each Decision has exactly one Macro Requirement whose concern is the natural semantic owner of the unresolved problem resolved by that Decision.**

Different MRs may mention the same domain objects and vocabulary, and a Decision may be relevant to other concerns. Relevance does not create multiple hierarchical ownership.

Unique-parent test: if a Decision can move under another MR without materially changing its Context or chosen meaning, review whether the Decision is too broad, redundant/derived, or whether MR boundaries fail to discriminate responsibility. A genuinely cross-cutting choice is not resolved by copying it under multiple MRs.

H-CLOSED-02 - Non-redundant Decision contribution

> **Each governed Decision contributes a significant choice that is not already owned by another applicable Decision and is not completely determined by its parent Macro Requirement plus the already governed choices.**

Two Decisions may share terminology, actors, consequences or evidence while owning different choices. The prohibited condition is duplicate semantic ownership of the same choice or promotion of a logically derived statement into a second Decision.

Decision-contribution test: if removing a candidate would not remove a significant project choice that cannot be reconstructed from the parent concern and remaining Decisions, review it as a possible duplicate, derived consequence, explanation or downstream obligation.

5 Three semantic validation levels

```
1. Local Decision validity
   semantic core / local problem / coherent chosen position

2. Hierarchical coherence
   one natural MR owner / discriminable parentage

3. Corpus coherence
   non-redundant contribution / no duplicate choice ownership
```

The second and third levels inherently require governed context beyond one file. Tooling may assist review at scale but does not own these semantics.

6 Rules deliberately relocated from this metamodel

Revision 7 retains the historical identifiers but changes their owner:

- **D-CLOSED-06 - No semantic patching**: authoring/review heuristic, not a Decision invariant. Non-goals remains outside the Decision semantic core.
- **D-CLOSED-08 - Lifecycle is not Decision-specific**: cross-cutting common-governance boundary. Lifecycle/status/history will be modeled separately.
- **D-CLOSED-09 - Semantic-owner separation**: cross-cutting semantic-authority principle applying to DDTA, governed projects and tools.

- **R-CLOSED-01..04**: representation and realization rules; see the model-layering foundation document.

Relocation is not rejection. It prevents rules about serialization or tool architecture from being treated as ontological properties of Decision.

7 Decision-to-downstream boundary - still partly OPEN

Working rule:

> **A Decision chooses; downstream governed documents make the chosen consequences operational and independently checkable.**

The exact semantics of the next Requirement level are not yet closed.

```
choice needed to explain what was adopted
    -> Decision

independently checkable obligation
    -> candidate next-level semantics

rollout / migration order
    -> planning

separable choice with its own trade-offs
    -> another Decision candidate
```

Technical vocabulary is not forbidden; the question is whether the detail belongs to the choice or to downstream realization.

8 Historical integrity

Decision Context may preserve circumstances true when the choice was taken. Later replacement should normally be represented through the future lifecycle/governance model rather than rewriting history to look current. Temporary rollout sequencing remains planning rather than Decision semantics.

9 Decision review tests after revision 7

The following tests assess the conceptual Decision model:

1. **Parent**: does the candidate narrow one parent MR?
2. **Unique parent**: is that MR the single natural semantic owner?
3. **MR discriminability**: if parentage is ambiguous, is the defect in the candidate or in MR boundaries?
4. **Contribution**: would removing it remove a significant choice not already governed or derivable?
5. **Decision-local Context**: does Context explain this unresolved choice rather than the whole MR?

6. **Explicit choice**: is an adopted position actually present?
7. **Coherent choice**: can the choice change as one conceptual unit?
8. **Consequences**: are material effects/trade-offs represented?
9. **Downstream boundary**: are independently checkable obligations descending rather than bloating the Decision?
10. **Historical integrity**: does Context preserve why the choice existed without carrying temporary rollout state?

Authoring heuristics, semantic-authority review, representation conformance and tooling checks are intentionally documented outside this Decision-specific test list.

10 Open Decision questions before candidate closure

1. Does re-review of the independent facial-access corpus expose a counterexample to the local core or H-CLOSED-01/H-CLOSED-02?
2. Where exactly is the threshold between precision intrinsic to a choice and next-level independently checkable detail?
3. Are consequence labels semantic categories or only authoring aids?
4. How should legitimate cross-MR relevance eventually be represented without multiple hierarchical ownership?

11 Validation sequence after layering correction

1. preserve the frozen facial-access D1 corpus;
2. preserve revision 7 as a layering checkpoint;
3. re-read the ThreatForge construction ADR regression without treating realization principles as metamodel semantics;
4. write a corrected candidate ADR set only after that regression is stable;
5. regress all prior evidence if Decision semantics themselves change;
6. only then proceed to sequential ThreatForge MR-0003/MR-0004/MR-0005 holdouts.

12 Explicitly deferred

- lifecycle/status value sets and transitions;
- Requirement-specific semantic model;
- cross-MR relevance relation, if needed;
- cross-document persistence/effective-context relation model;
- concrete representation mapping and serialization closure;
- concrete ThreatForge support/conformance relation model;
- concrete similarity/search algorithms and thresholds;
- ThreatForge migration/refactor.